TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Production of Knitted Fabric Samples Containing Basic Structures

Single jersey, 1×1 rib, interlock and purl knitted on the laboratory machines from one yarn, then measured on a common plan — so the structures themselves, not the materials, explain the differences in geometry, mass and behaviour.

FM-4202 · ADVANCED KNITTING SESSIONAL · EXPERIMENT 3 · FABRIC MANUFACTURING SPECIALIZATION

1. Aim and theory

Aim: to produce samples of the four basic weft-knitted structures from the same yarn at comparable stitch length, and to measure and compare their stitch geometry, areal mass and characteristic behaviour.

The four structures are the alphabet of weft knitting (TE-3101/TE-3205): **single jersey** (one bed, all loops one way), **1×1 rib** (two beds, wales alternating face and back), **interlock** (two intermeshed 1×1 ribs, two feeders per course), **purl** (loops alternating direction course by course). Holding yarn and stitch length constant isolates what the *structure* contributes: the GSM identity GSM = WPC × CPC × l × tex ÷ 100 (per bed-pair for rib/interlock, where each course carries two loop systems) then predicts each fabric's mass from its measured density — the experiment's central check.

2. Apparatus and materials

- Single-jersey circular machine (Experiment 1), rib/interlock-capable double-jersey machine and a purl-capable (double-ended needle / link-links) machine or flat machine as available; one yarn lot for all samples (24.6 tex cotton in the model run); stitch-length kit, GSM cutter and balance, measuring rule and pick glass, conditioning space.

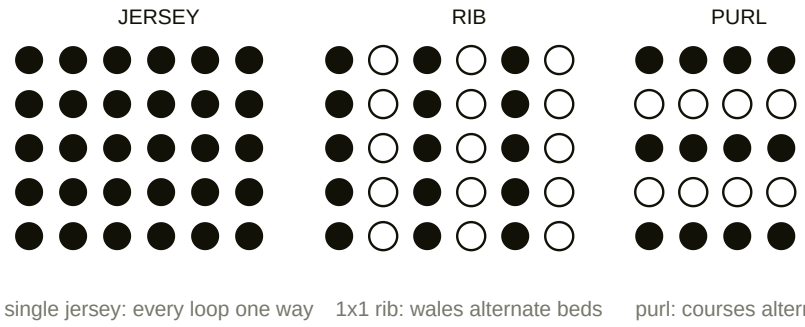


Figure 1 — Dot notation of the samples (convention of TE-3101/3205): filled dot = loop drawn to the face, open dot = loop drawn to the back. Jersey: one bed only. Rib: the beds alternate by wale. Purl: the direction alternates by course. Interlock (not drawn) is two such rib systems intermeshed back to back, so both faces read as jersey.

3. Procedure

1. Set each machine to the common yarn and adjust feeder/cam settings to deliver stitch length as near 3.2 mm as the structure allows; record the delivered l by the course-length method for every sample.
2. Knit a stable length of each structure (discard the setting-in metres); mark the state (on-machine / dry-relaxed) on each sample.
3. Condition samples flat, dry-relaxed, for 24 hours. Measure WPC and CPC (three positions each, pick glass), stitch length by unravelling a measured course, areal mass by GSM cutter, and thickness.
4. Handle tests: note curling (jersey), widthwise stretch and recovery (rib), firmness and face identity (interlock), lengthwise behaviour (purl); record spirality tendency on the jersey after a wash (developed fully in Experiment 5).
5. Test the GSM identity against each sample: compute GSM from WPC × CPC × l × tex (doubling the loop count per course for rib and interlock) and compare with the cutter value; explain residuals.

4. Worked calculation (model data)

THE IDENTITY CHECKED ON THE JERSEY SAMPLE

Jersey sample, dry-relaxed: WPC 11/cm, CPC 16/cm, l = 3.2 mm, yarn 24.6 tex.

GSM = 11 × 16 × 3.2 × 24.6 ÷ 100 = 176 × 3.2 × 24.6 ÷ 100 = 13,855 ÷ 100 ≈ **139 g/m²**.

Cutter value on the same sample: **140 g/m²** — agreement within 1%, the experiment's expected tolerance. (For rib and interlock the same arithmetic runs with the course's double loop system: their higher masses at equal l are arithmetic before they are anything else.)

5. Observation table (model data — one yarn, 24.6 tex)

STRUCTURE	WPC /CM	CPC /CM	L (MM)	GSM CALC / CUTTER	BEHAVIOUR NOTES
Single jersey	11	16	3.2	139 / 140	Curls to face at top/bottom, to back at sides; ladders readily
1×1 rib	10 (per face)	15	3.3	≈244 / 246	Great widthwise stretch and recovery; no curl; thicker
Interlock	11 (per face)	14 (per feeder pair)	3.1	≈235 / 236	Both faces jersey-like; firm, stable, least stretch of the set
Purl	11	13	3.3	≈116 / 117	Lengthwise stretch emphasis; both faces show sinker-loop arcs

Check the rib line as the examiner will: per full course the fabric carries two loop systems (one per bed): GSM = 2 × 10 × 15 × 3.3 × 24.6 ÷ 100 = 24,354 ÷ 100 ≈ 244 g/m². Interlock likewise doubles (its CPC counts feeder-pair courses).

6. Discussion points

- Same yarn, same l, masses from ≈116 to ≈246 g/m²: structure is a mass decision before it is a style decision.
- Jersey's curling and rib's stretch are the same fact — loop direction balance — read two ways (TE-3205).
- Why interlock is specified where stability matters (collars, uniforms): two intermeshed ribs lock each other's geometry.

7. Precautions

- Compare only conditioned samples: geometry measured on-machine or hot off it belongs to a different (unstated) relaxation state.
- Unravel stitch length from the same state as the density measurements, or the identity check compares two different fabrics.
- Mark each sample's structure, machine and state at the moment of cutting — unlabelled swatches are landfill.

8. Practice (viva preparation)

1. Why does interlock show a jersey face on both sides?

2. Using the identity, predict the GSM of a jersey knitted at WPC 12, CPC 17, l = 3.0 mm from 19.7 tex yarn.
3. A rib sample's cutter GSM is far above the single-system identity prediction. What was forgotten?
4. State the machine requirement that makes purl the rarest of the four in production.
5. Which sample would you specify for a cuff, and why — in structure terms?

Answer key

1. Interlock intermeshes two 1×1 rib systems back to back: each face is the face side of one rib system, so needle loops (the jersey face appearance) show on both sides, with the sinker loops locked inside.
2. GSM = 12 × 17 × 3.0 × 19.7 ÷ 100 = 204 × 3.0 × 19.7 ÷ 100 = 12,056 ÷ 100 ≈ **121 g/m²**.
3. The **second loop system**: a rib course carries loops on both beds — the identity must count both systems (factor 2), as worked in Section 5.
4. Purl needs loops transferred between beds course by course — double-ended (link-links) needles or transfer flat machines: slow and mechanically elaborate beside jersey/rib production, so it survives mainly as garment panels and design effects.
5. **1×1 rib**: its wales alternate beds, giving the set's best widthwise extension with recovery and no curl — exactly the cuff's duty of gripping without a separate elastic.
